

Micro Fluid Dispenser

Overview



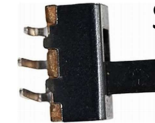
SMD Momentary Push switch – Start input



Slide Switch 1 – ON OFF



Slide Switch 2 – 30/60 day selection



Sensor 3 SMD Momentary Push switch – Refill Sensor



Motor 4.5V



Pinion



Sensor 2- Empty



Rack



AA Battery

AA Battery

AA Battery

Sensor 1 - Home



Logic / Sequence:

All ready Condition

1. New Refill is placed in position (Refill Sensor - Input)
2. Sensor 2 in NO
3. Sensor 1 in NC (Refill Full)

Push ON (Start input pulse) switch to start cycle. If we turn OFF in middle of the cycle using ON OFF slide switch, the cycle should continue from where it stopped by pressing the Start input pulse again (either internal memory or without cutting power to Controller IC irrespective of ON-OFF switch position)

Pre Pumping:

As soon as the start push button ON (momentary) – the motor will be on for a pre set time (max 9 seconds least count 0.1 seconds), provide 3 pre-pumping loop with common dealt. After a Pre-pumping a fixed delay D1 of 15 min

Pumping

After the delay D1, the pumping time starts (max settable time 1 second, least count 0.01 seconds)

Pumping interval D2 max 3 hrs least count 1 second

Pumping sequency will continue till the Sensor 2 becomes NC. On that moment the motor should start running in reverse and run till the sensor 1 becomes NC. At this point throw an indication to change the refill (refill Empty). During Revere running Different LED indication required

Once the refill sensor Input turns NO and Again NC the next cycle can be enabled with push button. LED indication to identify this condition

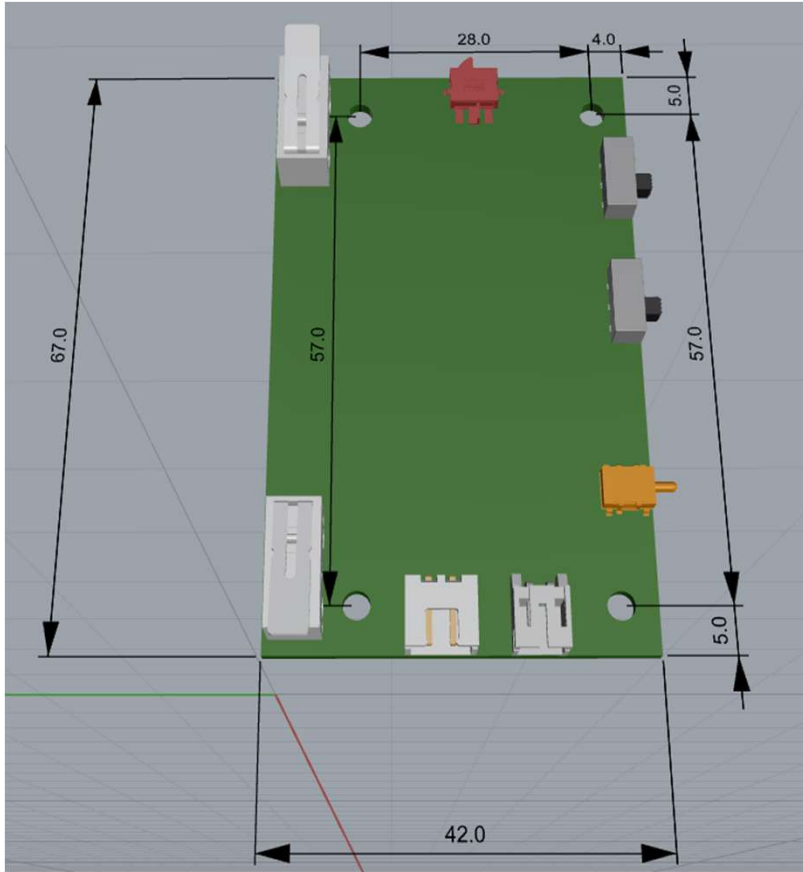
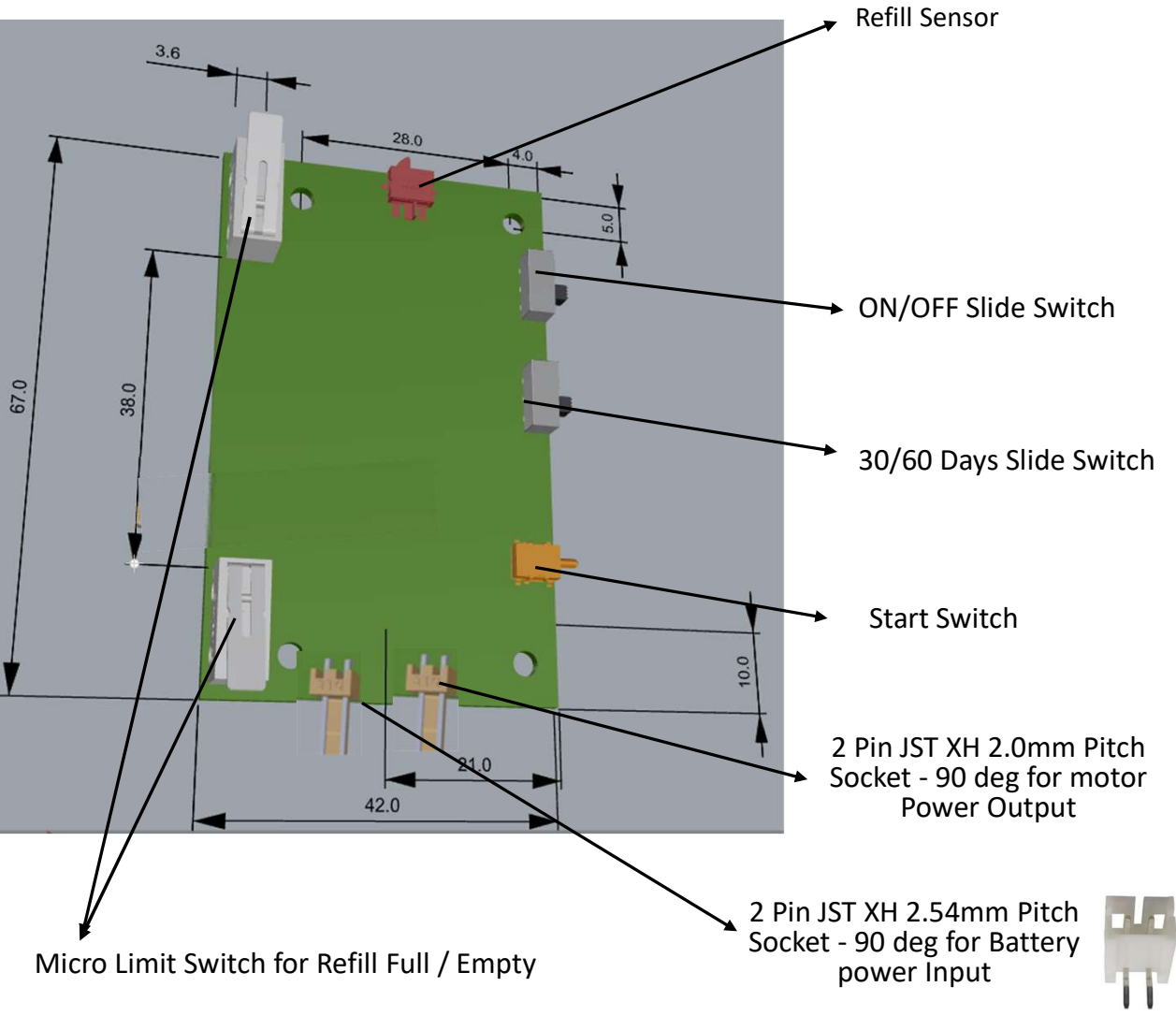
Battery voltage to be monitored and throw error in case voltage drops below 4.0V

Indications

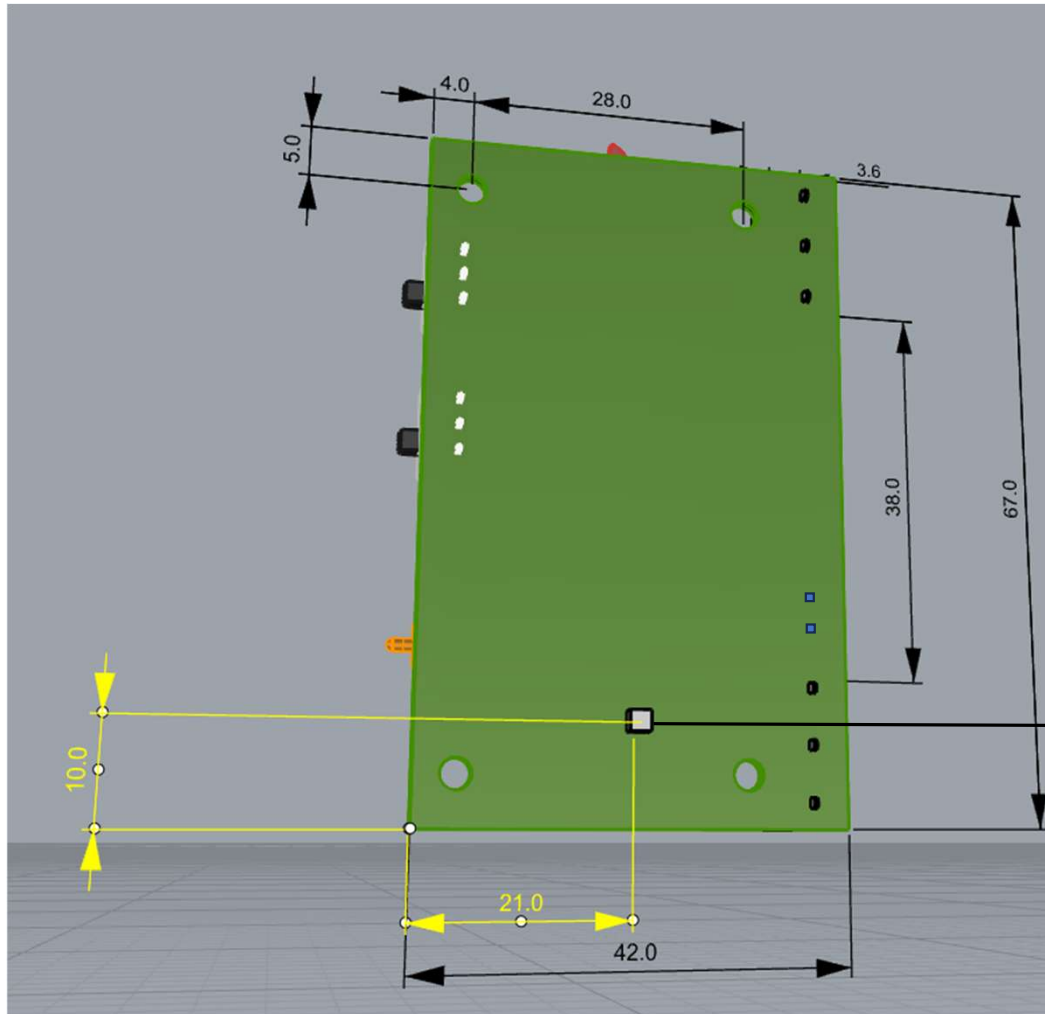
Sensor 1	Sensor 2	Sensor 3	State of Machine	Indication Pattern Requirement
NC *first time	NO*First time	NO	Install Refill	RED Blinking (may be different type)
NC	NO	NC (after installing new refill first time)	Ready for cycle	Blinking Green
NC (after few cycles it becomes NO)	NO	NC	After Start Input pulse - Machine running OK	Green LED as long as Motor ON (both pre-pump and Pump)
NO until NC	From the moment NO to NC ((after few cycles it becomes NO)	NC	Indication for Container Empty	Motor reverse BLUE Blinking
After full Reversing NC	NO	NC	Replace Refill	RED Blinking
NC	NO	NC- NO- NC	Refill replace ready for cycle	Blinking Green (may be different style)
			Battery Low	Continuous Red
After testing we can arrive at no of pumping for 30 and 60 days. Based on that we can give an indication that 75% or 80% of fluid is consumes and an alert for buy / keep replacement stock				Intermittent Red between Green for motor on

IF Outputs are limited then we can go for 2 colour indications

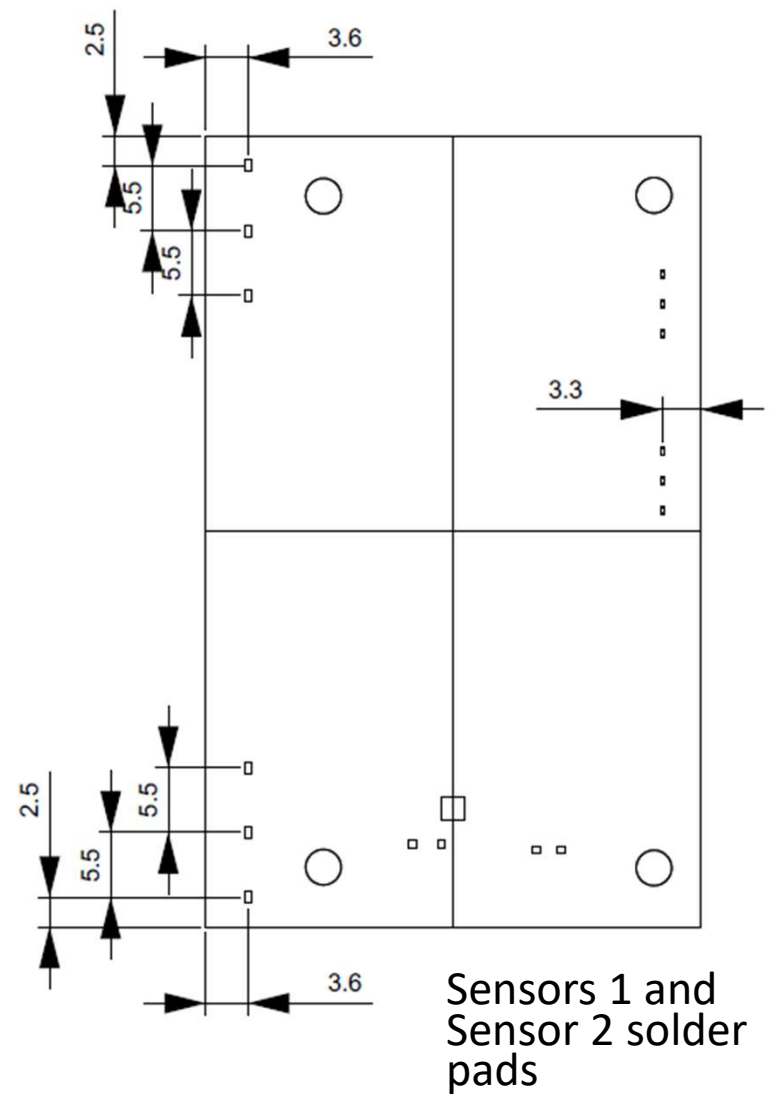
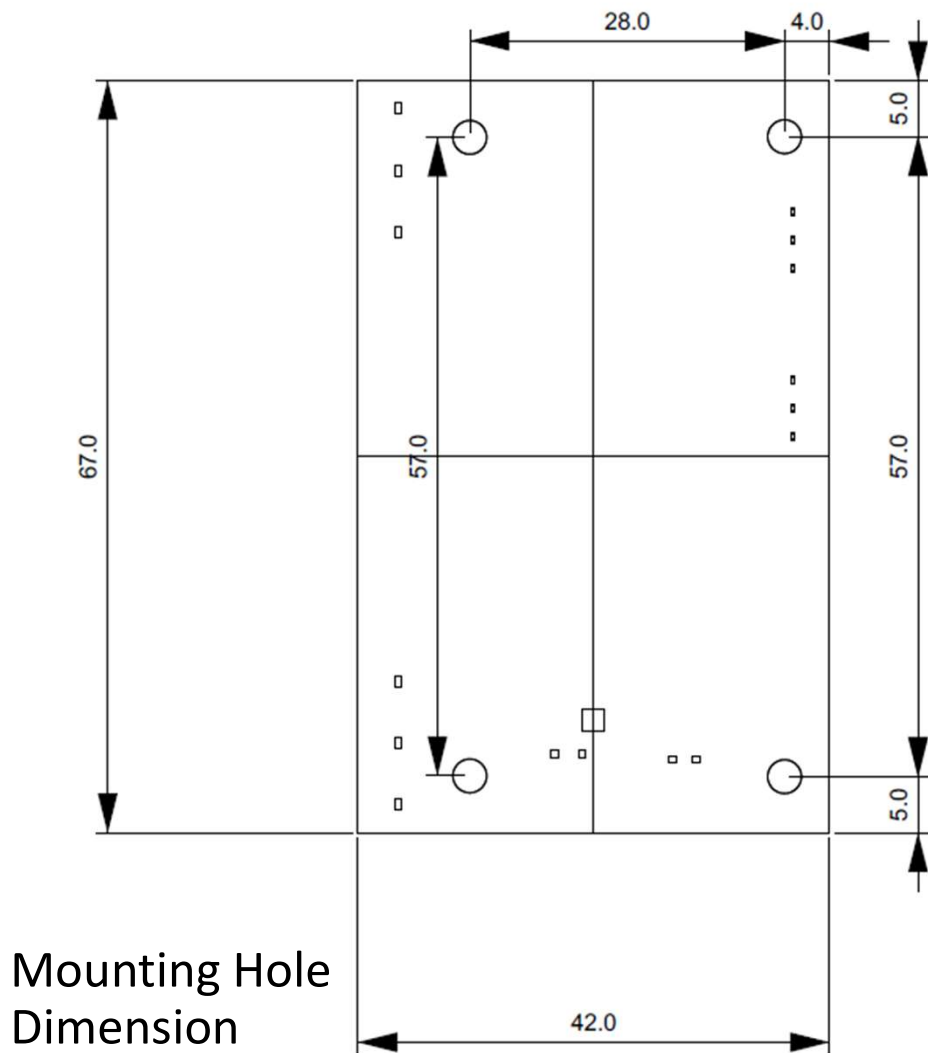
Dispenser PCB - Top

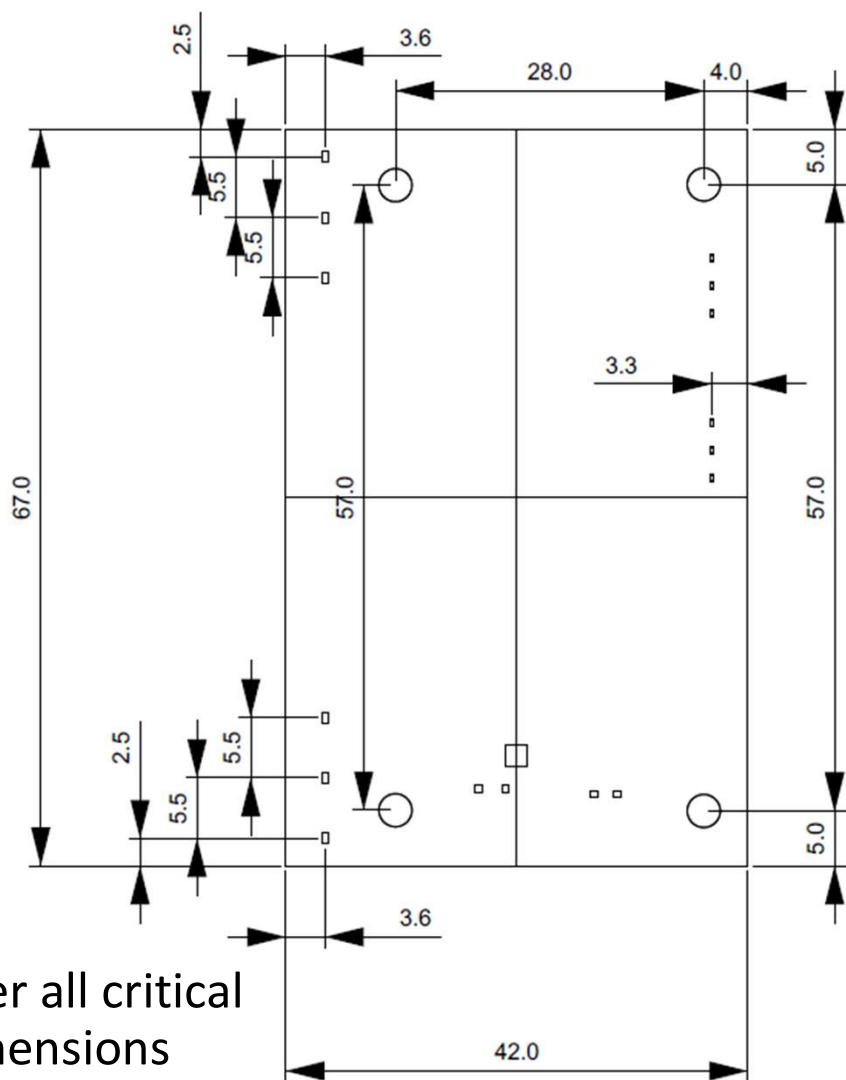


Dispenser PCB - Bottom

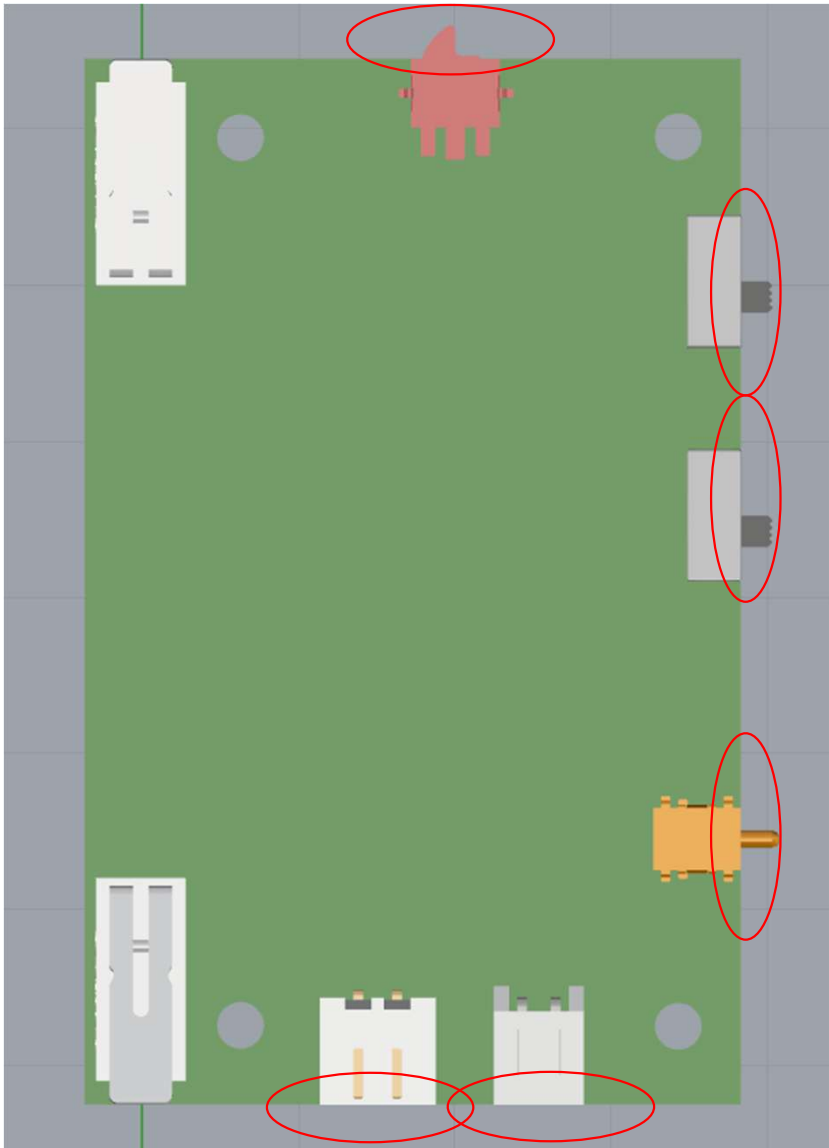


2/ 3 colour SMD LED it should be in bottom side but position can be changed if required





Over all critical
dimensions



Points to consider:

1. The mounting hole dimensions and over all PCB size are to maintained as per the drawing – Slide 7
2. The sensor 1 and Sensor 2 through hole dimensions to maintained as per drawing – Slide 7
3. Refill sensor, Slide ON-OFF switch, 30/60 day selector switch, Start Push switch, 2 pin JST for input power and 2 pin JST for Motor output power can be designed as per the convenience but all the components to be flushed to the edge of PCB

Components USED

Key Components used in Proto Development

(only for cost reference, Can suggest alternative considering cost / IO requirement)

1. ATTiny414-SS (14 pin SMD) - 8-bit Microcontrollers - MCU 20MHz, 4KB
2. DRV8837C - Motor Controllers & Drivers H-bridge – We faces issue of soldering due to 8-WSON Package.
3. ASDK2-32.768KHz-LRT (Standard Clock Oscillators AUTO-GRADE)
4. LED 3 Colour

Sensors:

1. Sensor 1 & 2 - KW10-DZ2P075-SPDT Long Lever Micro Switch-2A 125VAC(0.75N)
2. Sensor 3 - TM-1617SA-XKB-100mA 30V Double-pole single-throw (DPST) NC Detection Switch
3. Start Input TM-1022-B-U-XKB-SMD-6P,5x4mm Limit Switches ROHS
4. Slide Switch 1 & 2 - SS-12F21G5-Slide 1P2T Angled Switch

Bluetooth Module:

1. DA14531MOD-00F01002 - Bluetooth Modules - 802.15.1 Bluetooth Low Energy 5.1 Module with integrated ARM Cortex M0+, memories and peripherals 8 GPIOs

Motor:

1. Micro 300 Series Gear Motor DC3V-9V

Testing Interface

(testing to freeze all the timing)

Pre-Pumping 1	0.1 sec	UPDATE
Pre-Pumping 2	0.1 sec	
Pre-Pumping 3	0.1 sec	
Pre-Pumping Delay	1 sec	
Pumping Time	0.01 sec	
Pumping interval	0.1 sec	
No of Pumping	732	(count and show number of Pumping in full cycle here)
Running Status		Indicate the status here

Thank You!